Dillon Ma
Team 6
Teammates: Thomas Wang, Mithun Roy, Leena Aljurashi
ILR 2
Feb 12, 2026

Individual Progress

For this week's design proposal, I worked on the design requirements, the functional architecture, as well as the schedule for our project. For the functional architecture, I broke down the system's functions into its separate parts: power, mechanical support, calculations, controls, and locomotion. For the schedule of our team progress, I created a detailed gantt chart with all of the due dates and deliverables, and have planned out separately what systems should work together and when we should have those done by (ie. mechanical and electrical design need to work together to ensure proper wiring).

Separate from the presentation itself, I also created a simple CAD model of our design to demonstrate the size and functionality of our hexapod. The dimensions will be iterated upon and perfected based on more detailed calculations that need to be made. I also worked with Mithun and looked into possible motors that we want to use on our hexapod. For the design proposal, we landed on using 18 stepper motors, but it has been challenging to find motors that are both fast enough and have enough torque. We are considering possibly switching to DC motors and gearboxes.

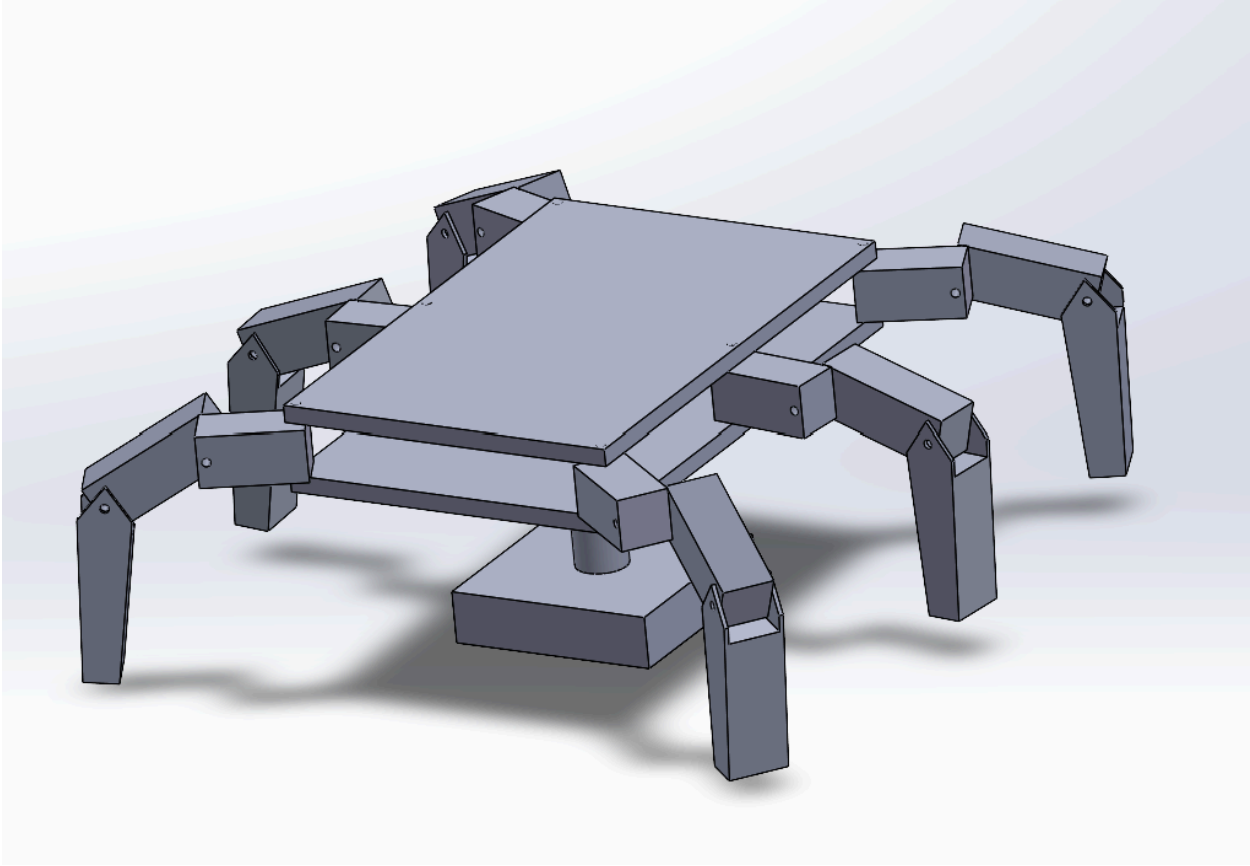


Figure 1: Preliminary simple Hexapod CAD model

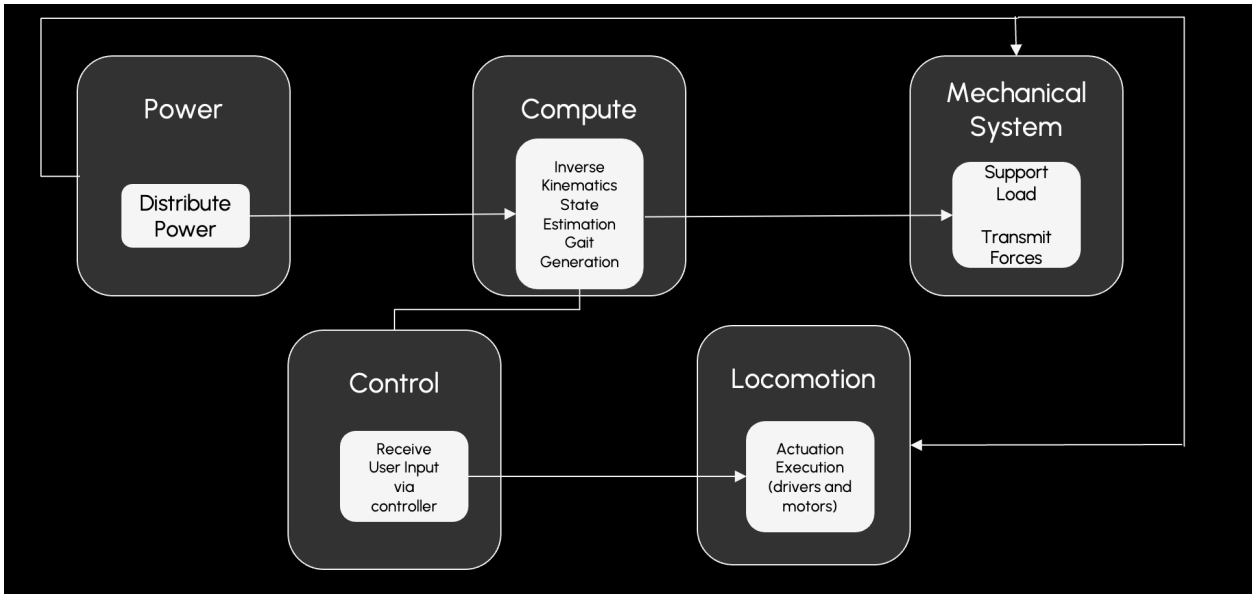
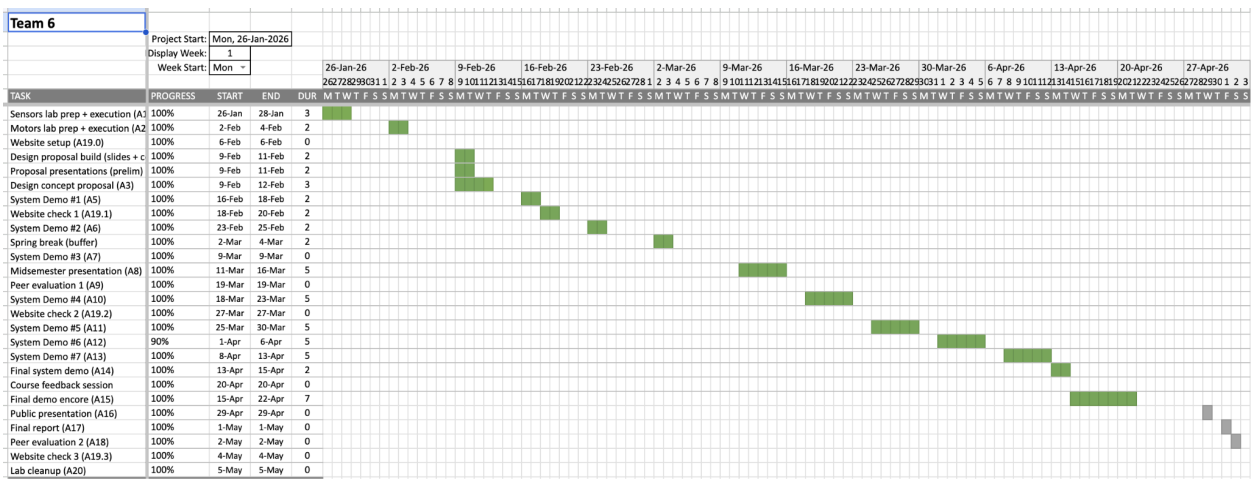


Figure 2: Functional Architecture of Hexapod



Team 6

Project Start: Mon, 26-Jan-2026
Display Week: 1
Week Start: Mon

TASK	PROGRESS	START	END	DUR
Sensors lab prep + execution (A1	100%	26-Jan	28-Jan	3
Motors lab prep + execution (A2	100%	2-Feb	4-Feb	2
Website setup (A19.0)	100%	6-Feb	6-Feb	0
Design proposal build (slides + c	100%	9-Feb	11-Feb	2
Proposal presentations (prelim)	100%	9-Feb	11-Feb	2
Design concept proposal (A3)	100%	9-Feb	12-Feb	3
System Demo #1 (A5)	100%	16-Feb	18-Feb	2
Website check 1 (A19.1)	100%	18-Feb	20-Feb	2
System Demo #2 (A6)	100%	23-Feb	25-Feb	2
Spring break (buffer)	100%	2-Mar	4-Mar	2
System Demo #3 (A7)	100%	9-Mar	9-Mar	0
Midsemester presentation (A8)	100%	11-Mar	16-Mar	5
Peer evaluation 1 (A9)	100%	19-Mar	19-Mar	0
System Demo #4 (A10)	100%	18-Mar	23-Mar	5
Website check 2 (A19.2)	100%	27-Mar	27-Mar	0
System Demo #5 (A11)	100%	25-Mar	30-Mar	5
System Demo #6 (A12)	90%	1-Apr	6-Apr	5
System Demo #7 (A13)	100%	8-Apr	13-Apr	5
Final system demo (A14)	100%	13-Apr	15-Apr	2
Course feedback session	100%	20-Apr	20-Apr	0
Final demo encore (A15)	100%	15-Apr	22-Apr	7
Public presentation (A16)	100%	29-Apr	29-Apr	0
Final report (A17)	100%	1-May	1-May	0
Peer evaluation 2 (A18)	100%	2-May	2-May	0
Website check 3 (A19.3)	100%	4-May	4-May	0
Lab cleanup (A20)	100%	5-May	5-May	0

Figure 3: Timeline of Project

Challenges

The challenges our team faced this week was trying to find motors that would fit our desired specs and budget, since we are planning for 3 degrees of freedom. The budget we have allows about $135 for each motor, but most servo motors we looked into at that price range are not strong enough for the purpose of moving the hexapod legs.

Teamwork

There was a significant amount of team work this week for the design proposal. For the design trade analysis, we held meetings as a group to discuss which components we wanted to use, including what types of motors, microcontrollers, buying or making our own controller for the hexapod, etc. We also discussed the overall design, debating between a hexagonal shaped

base or a rectangular base. We weighted each criteria for each decision, and the results of our discussion are shown in the design proposal.

Plans:

For next week, we have system demo 1. Our plan for the coming week is to have a fully flushed out CAD model with our desired leg joint lengths, which we will also calculate. We also plan on finalizing our motor choice and ordering and testing the sensors we need so that we can start prototyping and testing the functions we want. We as a team will finalize the design and the calculations in order to meet the size specifications of the project. I will be focusing on the CAD model and the gait calculations for the legs.